

CI/CD Pipeline

Inlämningsuppgift

Den här guiden beskriver hur man sätter upp en fullständig CI/CD-pipeline för ett ASP.NET Core Web API med Azure DevOps och Azure. Syftet är att automatisera bygge och driftsättning av applikationen så att ny kod deployar automatiskt till Azure varje gång ändringar pushas till master-branchen – utan manuella steg.

Guiden är uppdelad i fyra faser:

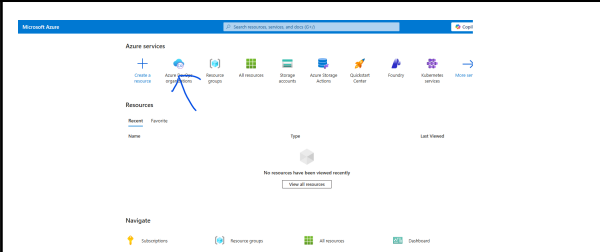
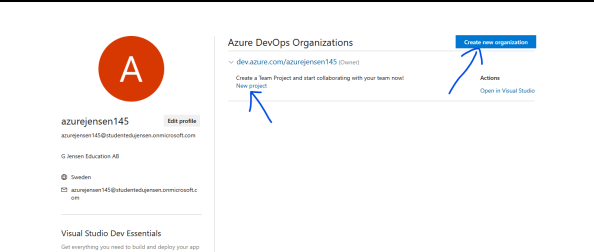
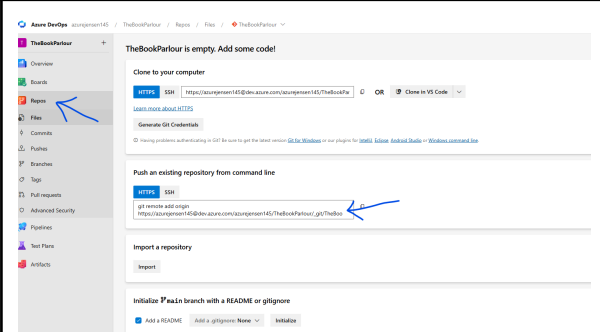
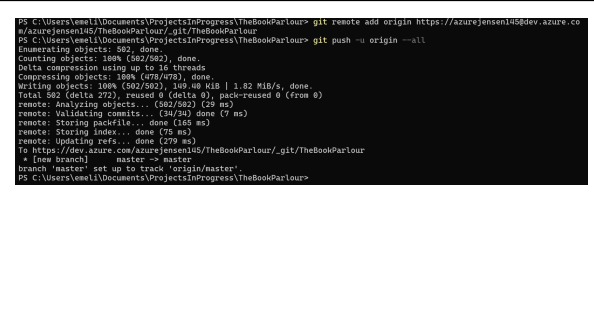
Fas 1 – Azure DevOps: koden laddas upp till ett Azure DevOps-repository.

Fas 2 – Azure: infrastrukturen skapas – resursgrupp, App Service, SQL-databas och Managed Identity konfigureras så att applikationen kan köras och nå databasen utan lösenord.

Fas 3 – Pipeline: en CI/CD-pipeline skapas i Azure DevOps med en YAML-fil som automatiskt bygger, publicerar och deployar applikationen vid varje push.

Fas 4 – Test: applikationen verifieras i produktion via Postman och eventuella brandväggsregler justeras.

Fas 1 – Azure DevOps (ladda upp koden)

 1. Logga in i Azure Portal och sök efter Azure DevOps	 2. Skapa en organisation och ett nytt projekt.
 3. Gå till Repos → kopiera koden under 'Push an existing repository from command line'	 4. Gå till projektmappen i Powershell och klistra in koden och kör kommandot.

Fas 2 – Azure (skapa infrastruktur)

```
C:\Users\jensl\Documents> git remote add origin https://azureservicebus.azure.com/C:\Users\jensl\Documents\Projects\Progress/TheBookParlour.git/TheBookParlour
C:\Users\jensl\Documents> git push --set-upstream origin main
branch master set up to track 'origin/master'
Everything up-to-date
C:\Users\jensl\Documents\Projects\Progress>TheBookParlour > $?sourceGroupname = "rg-TheBookParlour"
C:\Users\jensl\Documents\Projects\Progress>TheBookParlour > $location = "swedencentral"
C:\Users\jensl\Documents\Projects\Progress>TheBookParlour > az group create
    -Location $location
"id": "/subscriptions/9682709-57b1-4ea4-b092-17d9d36e5d0f/resourceGroups/rg-TheBookParlour",
"location": "swedencentral",
"managedBy": null,
"name": "rg-TheBookParlour",
"properties": {
  "provisioningState": "Succeeded"
}
"tags": null,
"type": "Microsoft.Resources.ResourceGroups"
```

1. Skapa resursgrupp

```
# Tilldela ett värde till en
variabel
$resourceGroupName =
"rg-TheBookParlour"
$location =
"swedencentral"

# Använd variabeln - $ används
vid BÅDE tilldelning och
användning
az group create `
--name $resourceGroupName `
--location $location
```

[illegible]

3. Skapa Webb App

```
az webapp create \
--name $webAppName \
--resource-group \
$resourceGroupName \
--plan $appServicePlanName \
--runtime "DOTNETCORE:10.0"
```

[illegible]

2. Skapa App Service Plan

```
$appServicePlanName =  
"asp-TheBookParlour"  
$webAppName         = "e-klint"  
  
az appservice plan create `--name $appServicePlanName `--resource-group  
$resourceGroupName `--location $location `--sku B1 `--is-linux
```

```

C:\Users\ekint\Documents\Projects\InProgress\TheBookParlour> sqlservr.exe -s "sql-ekint-ekint"
C:\Users\ekint\Documents\Projects\InProgress\TheBookParlour> sqlservr.exe -d "TheBookParlour"
C:\Users\ekint\Documents\Projects\InProgress\TheBookParlour> sqlservr.exe -u "sqladmin"
C:\Users\ekint\Documents\Projects\InProgress\TheBookParlour> sqlservr.exe -P "Secret1337"
C:\Users\ekint\Documents\Projects\InProgress\TheBookParlour> az sql server create --name sqlservr --resource-group ResourceGroupname --location Location --admin-user Administrator --admin-password AdminPassword

"administratorLogin": "sqladmin",
"administratorLoginPassword": null,
"administrators": null,
"createMode": null,
"externalGovernanceStatus": "Disabled",
"federatedLoginId": null,
"fullyQualifiedDomainName": "sql-ekint-ekint.database.windows.net",
"ids": ["/subscriptions/96587f9f-5701-46aa-b092-174903635001/resourceGroups/rg-TheBookParlour/providers/Microsoft.Sql/servers/sql-ekint"],
"identity": null,
"identityPrincipal": null,
"keyId": null,
"kind": "v12.0",
"location": "swedencentral",
"minimalTlsVersion": "1.2",
"name": "sql-ekint-ekint",
"primaryUserAssignedIdentityId": null,

```

4. Skapa logisk SQL Server

```
$sqlServerName      =  
"sql-api-eklint"  
$databaseName       =  
"TheBookParlourDb"  
$adminUser          = "sqladmin"  
$adminPassword      = "Secret#1337"
```

```
az sql server create \
  --name $sqlServerName \
  --resource-group \
  $resourceGroupName \
  --location $location \
  --admin-user $adminUser \
  --admin-password $adminPassword
```

```

PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az sql server show `
>> --name $sqlServerName `
>> --resource-group $resourceGroupName `
>> --query fullyQualifiedDomainName `
>> -o tsv
sql-api-eklint.database.windows.net
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> |

```

Svar:

sql-api-eklint.database.windows.net

Servern URL kommer att användas i steg 10, för att vi sedan ska kunna göra första migrationen.

5. Hämta sql-servers url

```

az sql server show `
--name $sqlServerName `
--resource-group $resourceGroupName `
--query fullyQualifiedDomainName `
-o tsv

```

```

PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az sql db create `
>> --resource-group $resourceGroupName `
>> --server $sqlServerName `
>> --name $databaseName `
>> --service-objective Basic
{
  "autoPauseDelay": null,
  "availabilityZone": "NoPreference",
  "catalogCollation": "SQL_Latin1_General_CP1_CI_AS",
  "collation": "SQL_Latin1_General_CP1_CI_AS",
  "createMode": null,
  "creationDate": "2026-05-19T09:27:45.238000+00:00",
  "currentBackupStorageRedundancy": "Geo",
  "currentServiceObjectiveName": "Basic",
  "currentSku": {
    "capacity": 5,
    "family": null,
    "name": "Basic",
    "size": null,
    "tier": "Basic"
  },
  "databaseId": "935c4ff7-9aff-4dc3-a18e-8f5ceb280009",
  "defaultSecondaryLocation": "swedensouth",
  "earliestRestoreDate": null,
  "edition": "Basic",
  "elasticPoolId": null,
  "elasticPoolName": null,
  "encryptionProtector": null,
  "encryptionProtectorAutoRotation": null,
  "failoverGroupId": null,

```

6. Skapa databas

```

az sql db create `
--resource-group $resourceGroupName `
--server $sqlServerName `
--name $databaseName `
--service-objective Basic

```

```

PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az sql server firewall-rule create `
>> --resource-group $resourceGroupName `
>> --server $sqlServerName `
>> --name AllowMyIP `
>> --start-ip-address 217.210.146.88 `
>> --end-ip-address 217.210.146.88
{
  "id": "/subscriptions/9662709-5741-4eaa-bb92-1749d3d654b0/resourceGroups/rg-TheBookParlour/providers/Microsoft.Sql/servers/sql-api-eklint/firewallRules/AllowMyIP",
  "name": "AllowMyIP",
  "resourceGroup": "rg-TheBookParlour",
  "startIpAddress": "217.210.146.88",
  "type": "Microsoft.Sql/servers/firewallRules"
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> |

```

```

PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az webapp identity assign `
>> --name $webAppName `
>> --resource-group $resourceGroupName
{
  "principalId": "4c2ed928-3645-4225-b7fb-75631bcb4599",
  "tenantId": "a49e271b-3af0-4d03-9e72-f61a5642e650",
  "type": "SystemAssigned",
  "userAssignedIdentities": null
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour>
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour>

```

Svar:

```

{
  "principalId": "4c2ed928-3645-4225-b7fb-75631bcb4599",
  "tenantId": "a49e271b-3af0-4d03-9e72-f61a5642e650",
  "type": "SystemAssigned",
  "userAssignedIdentities": null
}

```

7. Lägg till brandväggsregel

Som standard blockerar Azure SQL alla anslutningar. Vi måste explicit tillåta var IP (och senare Web App:ens IP i steg 3 under Fas 4- Test).

```

az sql server firewall-rule
create `
--resource-group $resourceGroupName `
--server $sqlServerName `
--name AllowMyIP `
--start-ip-address 217.210.146.88 `
--end-ip-address 217.210.146.88

```

8. Aktivera Managed Identity på web appen

```

az webapp identity assign `
--name $webAppName `
--resource-group $resourceGroupName

```

(Kör kommandot i terminalen i VS.)

9. Lägg till Azure connection string i User Secrets (tillfälligt)

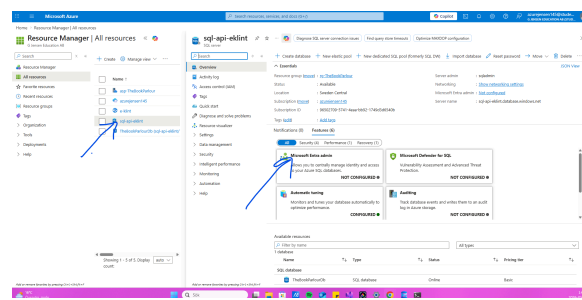
Använd serverns URL som vi fick i steg 5 för att kunna köra migrationen mot Azuredatabasen lokalt.. Glöm inte att byta tillbaka till din lokala connection string sedan.

```
dotnet user-secrets set
"ConnectionStrings:DefaultConnection"
"Server=tcp:sql-api-eklint.database.windows.net,1433;Database=TheBookParlourDb;Authentication=ActiveDirectory;TrustServerCertificate=True;Encrypt=True;"
```

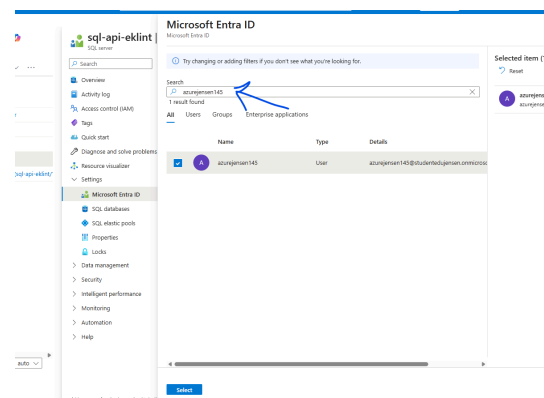
(Kör kommandot i terminalen i VS.)

10. Kör migration mot Azure-databasen

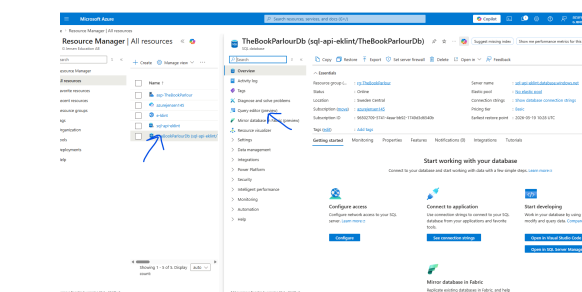
```
dotnet ef database update
```



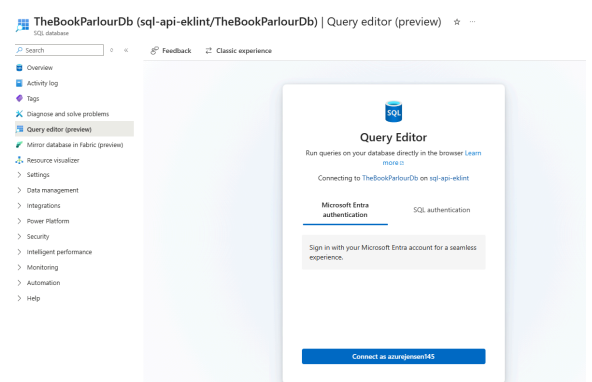
11. Gå till Azure Portalen och sök upp SQL Server. Välj Microsoft Entra Admin.



12. Sök upp ditt konto, bocka i det och klicka på **select**. Glöm inte att klicka på **Spara** innan du lämnar nästa sida.



13. Gå till SQL-databasen i portalen. Klicka på Query Editor.



14. Logga in med Microsoft EntraID authentication (skolkontot)

SQL query 1

Run Save as view

1 CREATE USER [e-klint] FROM EXTERNAL PROVIDER;

Messages Copy messages

3:11:41 PM Started executing on line 1
Affected rows: 0
Query executed successfully

15. Skapa databas användare kopplad till Managed Identity.

```
-- Skapa en databas användare  
kopplad till Managed Identity  
-- Namnet måste matcha exakt Web  
App-namnet i Azure  
CREATE USER [e-klint] FROM  
EXTERNAL PROVIDER;
```

SQL query 1

Run Save as view

1 ALTER ROLE db_datareader ADD MEMBER [e-klint];

Messages Copy messages

3:14:43 PM Started executing on line 1
Affected rows: 0
Query executed successfully

SQL query 1

Run Save as view

ALTER ROLE db_datawriter ADD MEMBER [e-klint];

Messages Copy messages

3:15:56 PM Started executing on line 1
Affected rows: 0
Query executed successfully

16. Ge databas användaren behörigheter.

```
-- Ge läsrättighet  
ALTER ROLE db_datareader ADD  
MEMBER [e-klint];  
  
-- Ge skriv rättighet  
ALTER ROLE db_datawriter ADD  
MEMBER [e-klint];  
  
-- Verifiera användaren  
SELECT name, type_desc,  
authentication_type_desc  
FROM sys.database_principals  
WHERE type IN ('S', 'E', 'X')  
ORDER BY name;
```

SQL query 1

Run Save as view

1 SELECT name, type_desc, authentication_type_desc
2 FROM sys.database_principals
3 WHERE type IN ('S', 'E', 'X')
4 ORDER BY name;

Messages Results Copy Download

	name	type_desc	authentication_type_desc
1	dbo	SQL_USER	INSTANCE
2	e-klint	EXTERNAL_USER	EXTERNAL
3	guest	SQL_USER	NONE
4	INFORMATION_SCHEMA	SQL_USER	NONE
5	sys	SQL_USER	NONE

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az webapp config appsettings set
--name $webAppName
--resource-group $resourceGroupName
--settings "ConnectionStrings__DefaultConnection=Server=tcp:sql-api-eklint.database.windows.net,1433;Database=TheBookParlourDb;Authentication=Active Directory Managed Identity;Encrypt=True;"
{
  "name": "ConnectionStrings__DefaultConnection",
  "slotSetting": false,
  "value": null
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

17. Sätt connection string i App Settings

Connection stringen innehåller bara adressen och säger "använd Managed Identity för autentisering" – inget lösenord alltså. Det är därför den är säker att lägga i App Settings!

```
az webapp config appsettings set
--name $webAppName
--resource-group $resourceGroupName
--settings
"ConnectionStrings__DefaultConnection=Server=tcp:sql-api-eklint.database.windows.net,1433;Database=TheBookParlourDb;Authentication=Active Directory Managed Identity;Encrypt=True;"
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az resource list -otable
Name Status ResourceGroup Location Type
-----
azure-jensen145 VisualStudioOnline-085779HFA0F201F68B82CAF0C4769HE westeurope microsoft.visualstudio/account Succeeded rg-TheBookParlour swedencentral Microsoft.Web/serve
app-TheBookParlour Succeeded rg-TheBookParlour swedencentral Microsoft.Web/sites
e-eklint Succeeded rg-TheBookParlour swedencentral Microsoft.Sql/serve
sql-api-eklint Succeeded rg-TheBookParlour swedencentral Microsoft.Sql/serve
ps Succeeded rg-TheBookParlour swedencentral Microsoft.Sql/serve
sql-api-eklint/master Succeeded rg-TheBookParlour swedencentral Microsoft.Sql/serve
rs/database Succeeded sql-api-eklint/TheBookParlourDb rg-TheBookParlour swedencentral Microsoft.Sql/serve
rs/database Succeeded PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

Det kan vara bra att lista resurserna för att få överblick och se att allt skapades korrekt.

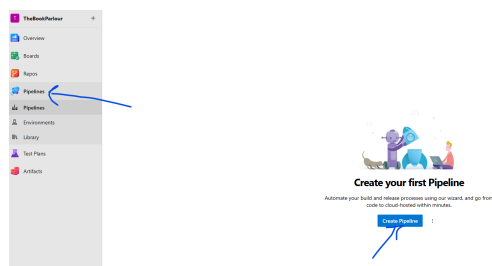
```
az resource list -otable
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az webapp identity show --name e-eklint --resource-group rg-TheBookParlour
{
  "principalId": "4c3ed928-3648-4225-b7fb-75631bcb4999",
  "tenantId": "449e272b-26f0-4082-b672-fc1a6a42eddd",
  "type": "SystemAssigned",
  "userAssignedIdentities": null
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

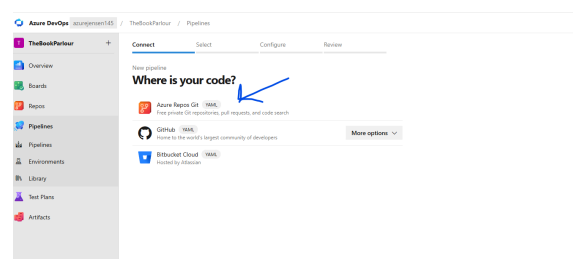
Även bra att kontrollera att **managed identity** är aktiverad.

```
az resource list -otable
```

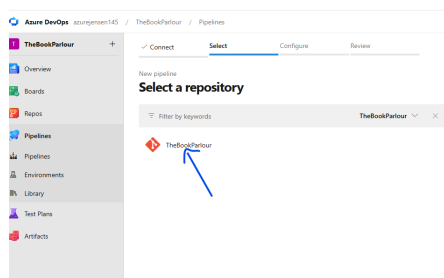
Fas 3 – Pipeline (koppla ihop allt)



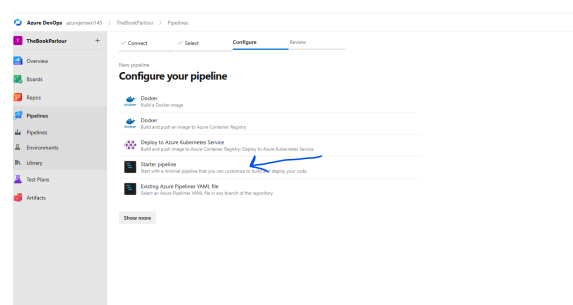
1. Gå till Azureportalen och projektet, klicka på Pipeline och "Create Pipeline".



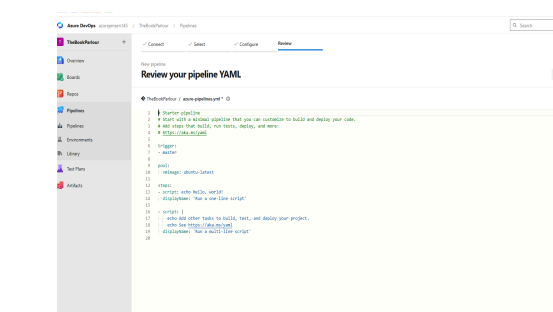
2. Gå dit koden finns – **Azure Repos Git** eftersom vi laddade upp koden där i fas 1.



3. Välj ditt repository i listan



4. Det finns flera olika sätt att konfigurera en pipeline, men jag valde att utgå från **Starter Pipeline** i listan.



5. Då skapas denna YAML-fil
 -pipelinen triggas när man pushar till **master**
 - Byggserver – Azure skapar en temporär virtuell maskin (**ubuntu-latest** som är linux och hoftast används för .NET Core och är plattformsoberoende)

Övriga steg behöver vi lägga in själva.

```
variables:
  buildConfiguration:
    'Release'

# Steg 1 - Installera beroenden (dotnet restore)
- task: DotNetCoreCLI@2
  displayName: Restore dependencies
  inputs:
    command: 'restore'
    projects: '**/*.slnx' # Hitta alla .slnx-filer rekursivt

# Steg 2 - Bygg koden
- task: DotNetCoreCLI@2
  displayName: Build project
  inputs:
    command: 'build'
    projects: '**/*.csproj'
    arguments: '--configuration Release'
```

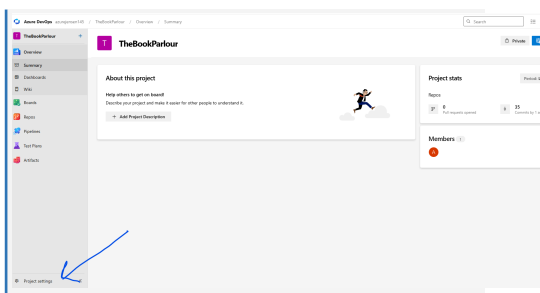
- 6.a) Definera variabeln \$(buildConfiguration) . Lägg till **tasks** för att installera beroenden och bygga koden.

```
# Steg 3 - Kör enhetstester
- task: DotNetCoreCLI@2
  displayName: Run tests
  inputs:
    command: 'test'
    projects:
      '**/*.UnitTests.csproj' # Bara
      testprojekt
    arguments: '--configuration
      Release --no-build'
```

6.b) För att enhetstesterna ska köras, bör man också lägga till tasken ovan.

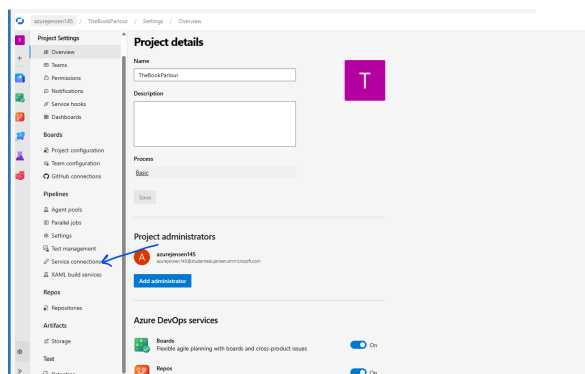
```
# Steg 4 - Publish
- task: DotNetCoreCLI@2
  displayName: 'Publish'
  inputs:
    command: 'publish'
    projects:
      'Backend/TheBookParlour.csproj '
    arguments: '--configuration
      $(buildConfiguration) --output
      $(Build.ArtifactStagingDirectory)
      /publish --no-build'
```

6.c) Lägg sedan till denna task för att publicera.

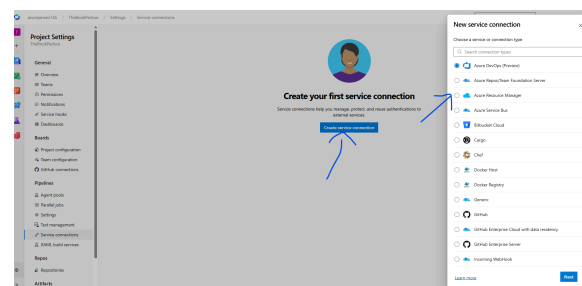


7. Gå till project settings

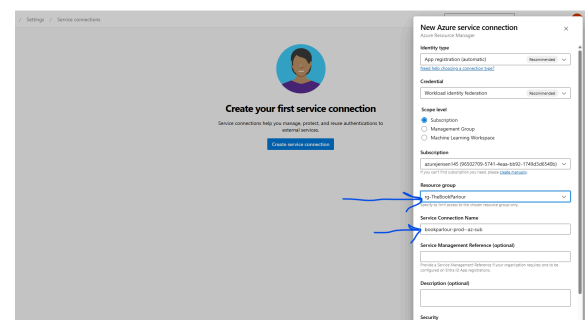
För att kunna skapa en "deploy task", måste vi först skapa vår **service connection** – en säker koppling mellan Azure DevOps och Azure.



8. Välj Service connections under Pipeline.



9. Välj Create Service Connection, sedan Azure Resource Manager och autentisera med ditt konto.



10. Välj resursgrupp och ge din service connection ett namn.

Jag bockade också i *Grant access permission to all pipelines*.

```
# 5 - Deploy till Azure
- task:
  AzureRmWebAppDeployment@5
    displayName: 'Deploy to
  Azure'
    inputs:
      ConnectionType: 'AzureRM'
      azureSubscription:
        'bookparlour-prod--az-sub'
      appType: 'webAppLinux'
      WebAppName: 'e-klint'
      packageForLinux:
        '$(Build.ArtifactStagingDirec
        tory)/publish/Backend.zip'
      RuntimeStack:
        'DOTNETCORE|10.0'
      DeploymentTypeLinux:
        'oneDeploy'
```

11. Lägg till denna task för att driftsätta.

(Fyll i din service connection, webb appens namn och vad zip-filens namn)

New pipeline

Review your pipeline YAML

◆ TheBookParlour / azure-pipelines.yml * 🔗

```
1 # Starter pipeline
2 # Start with a minimal pipeline that you can customize to build and deploy your code.
3 # Add steps that build, run tests, deploy, and more:
4 # https://aka.ms/pipelines
5
6 trigger:
7   - master
8
9 pool:
10  | vmImage: ubuntu-latest
11
12 variables:
13   - buildConfiguration: 'Release'
14
15 steps:
16
17 # Step 1 - Installa .NET Core (dotnet restore)
18
19 task: DotNetCoreCLI@2
20 displayName: Restore dependencies
21 inputs:
22   | command: 'restore'
23   | projects: '**/*.csproj' # Sätta alla .csn-filer rekursivt
24
25 # Step 2 - Bygg koden
26
27 task: DotNetCoreCLI@2
28 displayName: Build project
29 inputs:
30   | command: 'build'
31   | projects: '**/*.csproj'
32   | arguments: '--configuration Release'
33
34 # Step 3 - Här kan man lägga till enhetstester...
35
36 # Step 4 - Publish
37
38 task: DotNetCoreCLI@2
39 displayName: Publish
40 inputs:
41   | command: 'publish'
42   | arguments: '--configuration $(BuildConfiguration) --output $(Build.ArtifactStagingDirectory)/publish --no-build'
43
44 # Step 5 - Deploy till Azure
45
46 task: AzureWebAppDeployment@5
47 displayName: Deploy to Azure
48 inputs:
49   | ConnectionType: 'AzureRM'
50   | azureSubscription: 'bookparlour-prod--az-sub'
51   | appType: 'webAppLinux'
52   | packageForLinux: '$(Build.ArtifactStagingDirectory)/publish/TheBookParlour.zip'
53   | RuntimeStack: 'DOTNETCORE|10.0'
54   | DeploymentTypeLinux: 'oneDeploy'
```

12. När YAML-filen är komplett, tryck save and run högst upp i högra hörnet.

sen145 / TheBookParlour / Pipelines / TheBookParlour / 20260522.2

Jobs in run #20260...
TheBookParlour

Jobs

- Job 31s
- Initialize job 5s
- Checkout TheBo... 1s
- Restore depende... 9s
- Build project 10s
- Run tests <1s
- Publish 3s

Publish

```
1 Starting: Publish
2 =====
3 Task       : .NET Core
4 Description : Build, test, package, or publish a dotnet application, or run a custom command
5 Version     : 2.209.0
6 Author      : Microsoft Corporation
7 Help        : https://docs.microsoft.com/azure/devops/pipelines/tasks/build/dotnet-core
8 =====
9 Info: .NET Core SDK/runtime 2.2 and 3.0 are now end of life (EOL) and have been removed
10 /usr/bin/dotnet publish /home/vsts/work/1/s/Backend/TheBookParlour.csproj -f:Centrali
11 TheBookParlour -> /home/vsts/work/1/s/publish/Backend/
```

13. Sedan kan man gå till Pipelines och följa processen.

Azure DevOps azurejensen145 / TheBookParlour / Pipelines / TheBookParlour / 20260520.3

TheBookParlour

Overview

Boards

Repos

Pipelines

Environments

Library

Test Plans

Artifacts

Project settings

Jobs in run #20260...
TheBookParlour

Jobs

- Job 2m 26s
- Initialize job 4s
- Checkout TheBo... 3s
- Restore depend... 15s
- Build project 11s
- Publish 2s
- Deploy to A... 1m 48s
- Post-job Check... <1s
- Finalize Job <1s
- Report build sta... <1s

Report build status

1 Set build status for commit 090b043.

14. Processen klar!

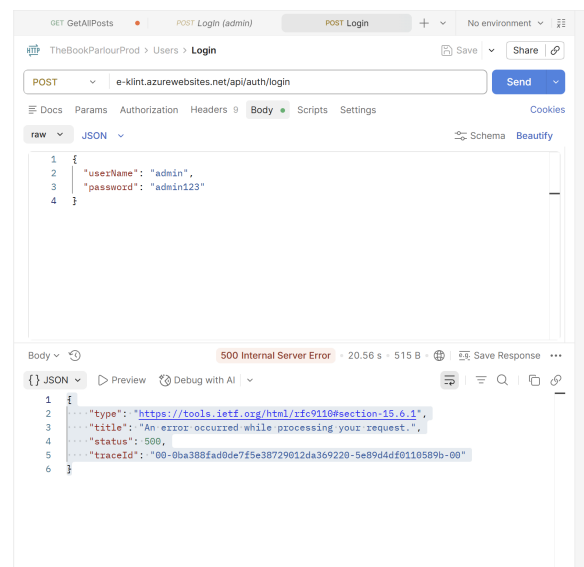
Fas 4 – Test (Kontrollera att Api:et fungerar i produktion)

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az webapp show `
> --name $webAppName `
> --resource-group $resourceGroupName `
> --query defaultHostName `
> -o tsv
e-klint.azurewebsites.net
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour>
```

Svar: e-klint.azurewebsites.net

1. Ta reda på webbappens URL

```
az webapp show --name $webAppName `
--resource-group $resourceGroupName --query
defaultHostName -o tsv
```



2. Testa en metod i postman, logga in

//Detta förväntas INTE att fungera eftersom brandväggen ännu inte tillåter webbappens IP-adress.

```
2026-05-20T10:48:48 Welcome, you are now connected to log-streaming service.
Starting Log Tail - n 10 of existing logs ---
/Support/Tools/Logs/runtime/containers/log
2026-05-20T10:47:13 43637062 at Microsoft.EntityFrameworkCore.Storage.RelationalCommand.ExecuteReaderAsync(RelationalCommandParameterObject parameterObject, CancellationToken cancellationToken)
2026-05-20T10:47:13 43637092 at Microsoft.EntityFrameworkCore.Query.Internal.SingleQueryingEnumerable`1.AsyncEnumerator.InitializeReaderAsync(AsyncEnumerator enumerator, CancellationToken cancellationToken)
2026-05-20T10:47:13 43637092 at Microsoft.EntityFrameworkCore.SqlServer.Storage.Internal.SqlServerExecutionStrategy.ExecuteAsync[TState,TResult](TState state, Func`4 operation, Func`4 verifySucceeded, CancellationToken cancellationToken)
2026-05-20T10:47:13 43638142 at Microsoft.EntityFrameworkCore.Query.Internal.SingleQueryingEnumerable`1.AsyncEnumerator.MoveNextAsync()
2026-05-20T10:47:13 43638222 ClientConnectionId: fa508dce-c564-41ac-893f-9778a7a3d580
2026-05-20T10:47:13 43638682 Error Number: 48615, State: 1, Class: 14
2026-05-20T10:47:13 43638672 ClientConnectionId: 70adcb66-faf6-4685-abb5-363a8dc9f57b
2026-05-20T10:47:13 43639052 Routing Destination: da6cd7363a8.tr2904.suedcentral1-a.worker.database.windows.net, 11018
2026-05-20T10:47:13 45785662 fail: TheBookParlour.Controllers.AuthController[500]
2026-05-20T10:47:13 45785672 ClientConnectionId: 70adcb66-faf6-4685-abb5-363a8dc9f57b
2026-05-20T10:47:13 45785672 135.116.238.63 is not allowed to access the server. To enable access, use the Azure Management Portal or run 'az sql server firewall-rule create --resource-group $resourceGroupName --server $sqlServerName --name AllowWebApp --start-ip-address 135.116.238.63 --end-ip-address 135.116.238.63'
2026-05-20T10:48:48 No new trace in the past 1 min(s).
2026-05-20T10:49:48 No new trace in the past 2 min(s).
2026-05-20T10:51:48 No new trace in the past 3 min(s).
2026-05-20T10:52:48 No new trace in the past 4 min(s).
2026-05-20T10:53:48 No new trace in the past 5 min(s).
```

Svar: Client with IP address '135.116.238.63' is not allowed to access the server.

3. Kör detta kommando för att hitta webb appens IP i loggen.

```
az webapp log tail --name $webAppName --resource-group $resourceGroupName
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour> az sql server firewall-rule create `
--resource-group $resourceGroupName `
--server $sqlServerName `
--name AllowWebApp `
--start-ip-address 135.116.238.63 `
--end-ip-address 135.116.238.63
{
  "endIpAddress": "135.116.238.63",
  "id": "/subscriptions/96027909-7f02-40a9-b092-170d1d6540b0/resourceGroups/rg-TheBookParlour/providers/Microsoft.Sql/servers/sql-klint/firewallRules/AllowWebApp",
  "name": "AllowWebApp",
  "resourceGroup": "rg-TheBookParlour",
  "startIpAddress": "135.116.238.63",
  "type": "Microsoft.Sql/servers/firewallRules"
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour>
```

```
az sql server firewall-rule
create `
--resource-group
$resourceGroupName `
--server $sqlServerName `
--name AllowWebApp `
--start-ip-address
135.116.238.63 `
--end-ip-address 135.116.238.63
```

4. Lägg till brandväggsregel för webbappens IP-adress.

Obs! Viktigt att byta namn på regeln, så den inte skriver över regeln i Fas 2 -steg 7.



Azure Key Vault

Obs! Denna guide förutsätter att du redan har en resursgrupp, app service plan och webbapp från CI/CD Guiden (Fas 2, steg 1-3).

Fas 1: Skapa Azure Key Vault och lägga till hemlighet (jwt-key)

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> $resourceGroupName = "rg-TheBookParlour"
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> $location = "swedencentral"
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> $keyVaultName = "kv-lab-e-klint"
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

1. Sätt variabler

```
$resourceGroupName =
"rg-TheBookParlour"
$location =
"swedencentral"
$keyVaultName =
"kv-lab-e-klint"
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault create `
--name $keyVaultName `
--resource-group $resourceGroupName `
--location $location `
--enable-rbac-authorization true
{
  "id": "/subscriptions/96502789-5741-4eaa-bb92-1749d3d6540b/resourceGroups/rg-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-klint",
  "location": "swedencentral",
  "name": "kv-lab-e-klint",
  "properties": {
    "accessPolicies": [],
    "createMode": null,
    "enableRbacAuthorization": true,
    "enableSoftDelete": true,
    "enableSoftDelete": true,
    "enableSoftDelete": false,
    "enableSoftDelete": false,
    "enableSoftDelete": false,
    "enableSoftDelete": false,
    "networkAcls": null,
    "privateEndpointConnections": null,
    "provisioningState": "Succeeded",
    "publicNetworkAccess": "Enabled",
    "sku": {
      "family": "A4",
      "name": "standard"
    },
    "softDeleteRetentionInDays": 90
  }
}
```

2. Skapa Key Vault

Flaggan `--enable-rbac-authorization true` aktiverar RBAC-modellen (rekommenderas framför den äldre access policy-modellen).

```
az keyvault create `
--name $keyVaultName `
--resource-group $resourceGroupName `
--location $location `
--enable-rbac-authorization true
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> $userId = az ad signed-in-user show --query id -o tsv
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> $userId = az ad signed-in-user show --query id -o tsv
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az role assignment create `
--assignee-object-id $userId `
--assignee-principal-type User `
--role "Key Vault Secrets Officer" `
--scope $keyVaultId
{
  "condition": null,
  "conditionVersion": null,
  "createdOn": "2026-05-22T12:23:18.150477+00:00",
  "createdOn": null,
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/96502789-5741-4eaa-bb92-1749d3d6540b/resourceGroups/rg-TheBookParlour/providers/Microsoft.Authorization/roleAssignments/01ba878b-2fdd-4ddf-9684-74d629804282",
  "name": "01ba878b-2fdd-4ddf-9684-74d629804282",
  "principalId": "8a307c2b-7fc6-461d-9899-311254d6162",
  "principalType": "User",
  "resourceGroup": "rg-TheBookParlour",
  "roleDefinitionId": "/subscriptions/96502789-5741-4eaa-bb92-1749d3d6540b/providers/Microsoft.Authorization/roleDefinitions/0ba878b-2fdd-4ddf-9684-74d629804282",
  "scope": "/subscriptions/96502789-5741-4eaa-bb92-1749d3d6540b/resourceGroups/rg-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-klint",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "8a307c2b-7fc6-461d-9899-311254d6162",
  "updatedAt": "2026-05-22T12:23:18.122041+00:00"
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

3. Hämta ditt entra-ID och Key Vault-ID.

Ditt användar-ID i Entra ID

```
$userId = az ad signed-in-user
show --query id -o tsv

# Key Vault:ets resurs-ID (behövs
för att sätta scope på roller)
$keyVaultId = az keyvault show `
--name $keyVaultName `
--query id `
-o tsv
```

4. Tilldela dig själv rollen som Secrets Officer

```
az role assignment create `
--assignee-object-id $userId `
--assignee-principal-type User `
--role "Key Vault Secrets Officer" `
--scope $keyVaultId
```



```

PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend> $jwtKey = "ThisIsAVerySecretJwtKey_ForBookshop!"
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend>
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault secret set ' ' --vault-name $keyVaultName
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault secret set ' ' --name "Jwt--Key" --value $jwtKey
{
  "attributes": {
    "created": "2026-05-22T13:09:19+00:00",
    "enabled": true,
    "expires": null,
    "notBefore": null,
    "recoverableDays": 90,
    "recoveryLevel": "Recoverable+Purgeable",
    "updated": "2026-05-22T13:09:19+00:00"
  },
  "contentType": null,
  "id": "https://kv-lab-e-kLint.vault.azure.net/secrets/Jwt--Key/87d8551ef27ucf3b328bc0e1f9a19e",
  "kid": null,
  "managed": null,
  "name": "Jwt--Key",
  "tags": {
    "file-encoding": "utf-8"
  },
  "value": "ThisIsAVerySecretJwtKey_ForBookshop!"
}
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend>

```

5. Sätt variabel och lägg till hemlighet.

```

$jwtKey =
"ThisIsAVerySecretJwtKey_ForBooks
hop!"

az keyvault secret set `
--vault-name $keyVaultName `
--name "Jwt--Key" ` --value
$jwtKey

```

```

PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault secret show `
>> --vault-name $keyVaultName `
>> --name "Jwt--Key" `
>> --query value
>> -o tsv
ThisIsAVerySecretJwtKey_ForBookshop!
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend>

```

6. Hämta heligheten

För att kontrollera att hemligheten lagrades korrekt i Key Vault, att du har rättigheter att läsa den samt att värdet är korrekt.

```

az keyvault secret show `
--vault-name $keyVaultName `
--name "Jwt--Key" ` --query value
-o tsv

```

```

PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend> az role assignment list `
>> --assignee $userId `
>> --scope $keyVaultId `
>> -o table
Principal                                Role                                Scope
-----
azurejensen105@studnetedu.jensen.nmicrosoft.com Key Vault Secrets Officer /subscriptions/95502709-5741-4eaa-b092-1749d
3d6840b/resourceGroups/rg-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-kLint
PS C:\Users\emelie\Documents\ProjectsInProgress\TheBookParlour\Backend>

```

7. Verifiera dina roller

```

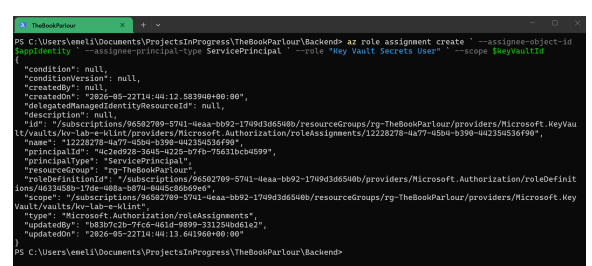
az role assignment list `
--assignee $userId `
--scope $keyVaultId `
-o table

```

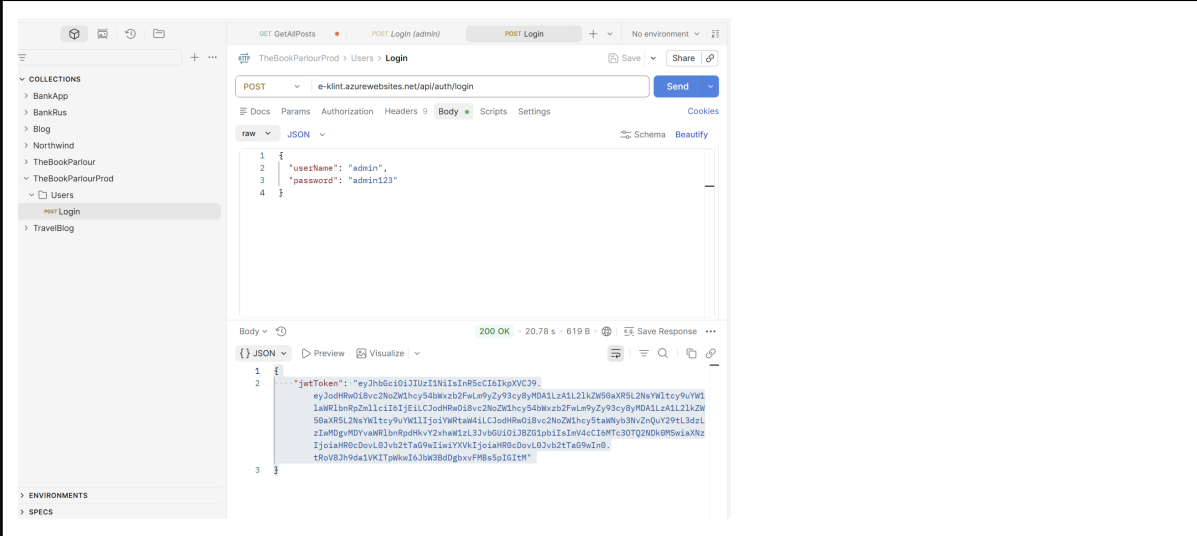
Fas 2: Integrera Key Vault i .NET-applikationen

1. Installera nuget-paket i Visual Studio (Package Manager Console) <pre>dotnet add package Azure.Identity dotnet add package Azure.Extensions.AspNetCore.Configuration.Secrets</pre>	2. Lägg till i program <pre>builder.Configuration.AddAzureKeyVault(new Uri("https://kv-lab-e-klint.vault.azure.net/"), new DefaultAzureCredential());</pre> <i>Din Key Vault URI följer alltid samma format: https://{ditt-keyvault-namn}.vault.azure.net/</i>
3. Se till att nyckeln hämtas från konfigurationen i metoden som genererar tecken. <pre>public class TokenService : ITokenService { private readonly IConfiguration _configuration; public TokenService(IConfiguration configuration) { _configuration = configuration; } public string GenerateToken(User user) { // Hämta nyckeln från Key Vault via IConfiguration var jwtKey = _configuration["Jwt:Key"]; var key = new SymmetricSecurityKey(Encoding.UTF8. GetBytes(jwtKey)); // Forts. på metoden.... } }</pre>	Obs! Se till att DI är injicerat i program också! <pre>builder.Services.AddScoped<ITokenService, TokenService>();</pre>

Fas 3: Ge App Service tillgång till Key Vault (med managed identity)

 <pre>PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> \$appIdentity = az webapp identity show --name \$webAppName --resource-group \$resourceGroupName --query principalId --output tsv PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> </pre>	 <pre>PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az role assignment create --assignee-object-id \$appIdentity --assignee-principal-type ServicePrincipal --role "Key Vault Secrets User" --scope \$keyVaultId { "condition": null, "conditionVersion": null, "createdBy": null, "createdOn": "2026-05-22T14:44:12.583940+00:00", "delegatedManagedIdentityResourceId": null, "id": "/subscriptions/96502709-6741-4eaa-bb92-1709d3d6540b/resourceGroups/rp-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-kint/providers/Microsoft.Authorization/roleAssignments/12228278-4a77-4504-b398-442354536f90", "name": "12228278-4a77-4504-b398-442354536f90", "principalId": "4c2ed928-3645-4228-0748-75631bcb4599", "principalType": "ServicePrincipal", "resourceGroup": "rp-TheBookParlour", "roleDefinitionId": "/subscriptions/96502709-6741-4eaa-bb92-1709d3d6540b/providers/Microsoft.Authorization/roleDefinitions/4d33458b-17de-488a-b074-b6e5c86d694d", "scope": "/subscriptions/96502709-6741-4eaa-bb92-1709d3d6540b/resourceGroups/rp-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-kint", "type": "Microsoft.Authorization/roleAssignments", "updatedBy": "4830fc3b-7f6c-b61e-8099-13125a0d142", "updatedOn": "2026-05-22T14:44:13.641940+00:00" }</pre>
1. Hämta App Service identity <pre>\$appIdentity = az webapp identity show --name \$webAppName --resource-group \$resourceGroupName --query principalId --output tsv</pre>	2. Ge rättigheter <pre>az role assignment create --assignee-object-id \$appIdentity --assignee-principal-type ServicePrincipal --role "Key Vault Secrets User" --scope \$keyVaultId</pre>

Fas 4: Kontrollera att metod med jwt fungerar i postman

 The screenshot shows a Postman collection named 'TheBookParlour' with a folder 'Users' containing a 'Login' endpoint. The request is a POST to 'e-klint.azurewebsites.net/api/auth/login' with a JSON body: { "username": "admin", "password": "admin123" }. The response is a 200 OK with a JSON body containing a 'jwtToken'.	Klart!
---	----------------------

Application Insights

Obs! Denna guide förutsätter att du redan har en resursgrupp, app service plan och webbapp från CI/CD Guiden (Fas 2, steg 1-3). Den förutsätter också att du har satt upp ett Azure Key Vault för att lagra hemligheter.

Fas 1: Sätta upp Applications Insight i Azure

```
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $resourceGroupName = "rg-TheBookParlour"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $location = "swedencentral"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $appServicePlanName = "asp-TheBookParlour"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $webAppName = "e-klint"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $appInsights = "appi-eklint"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $workspace = "eklint-lab"
```

1. Sätt miljövariabler

```
$appInsights = "appi-eklint"
$workspace = "eklint-lab"
$resourceGroupName = "rg-TheBookParlour"
$location = "swedencentral"
$appServicePlanName = "asp-TheBookParlour"
$webAppName = "e-klint"
```

```
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $resourceGroupName = "rg-TheBookParlour"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $location = "swedencentral"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $appServicePlanName = "asp-TheBookParlour"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $webAppName = "e-klint"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $appInsights = "appi-eklint"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $workspace = "eklint-lab"
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> az monitor log-analytics workspace create --resource-group $resourceGroupName --location $location --workspace $workspace
Resource provider 'Microsoft.OperationalInsights' used by this operation is not registered. We are registering for you.
Registration succeeded.
{
  "createdDate": "2026-05-22T10:04:04.412848Z",
  "customerId": "d229a8a8-7272-7f46-a9b6-8c92ec6486dc",
  "etag": "\"1a1002f0-0000-7800-0000-6a102a10000\"",
  "features": {
    "enableLogAccessUsingOnlyResourcePermissions": true,
    "legacy": 0,
    "searchVersion": 1
  },
  "id": "/subscriptions/66502789-5701-4eaa-b092-17b9d3d650b0/resourceGroups/rg-TheBookParlour/providers/Microsoft.OperationalInsights/workspaces/eklint-lab",
  "location": "swedencentral",
  "modifiedDate": "2026-05-22T10:04:17.302967Z",
  "name": "eklint-lab",
  "provisioningState": "Succeeded",
  "type": "Microsoft.OperationalInsights/workspaces"
}
```

2. Skapa Log analytics Workspace

```
az monitor log-analytics workspace create \
  --workspace-name $workspace \
  --resource-group $resourceGroupName \
  --location $location
```

```
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> az monitor app-insights component create --app $appInsights --location $location --resource-group $resourceGroupName
Please run 'az config set extension.dynamic_install_allow_preview=true or false' to config it specifically.
The command requires the extension application-insights. Do you want to install it now? The command will continue to run after the extension is installed. (Y/n)
Run 'az config set extension.use_dynamic_install_without_prompt' to allow installing extensions without prompt.
Extension 'application-insights' has a later preview version to install, add --allow-preview true to try preview version.
Resource provider 'Microsoft.OperationalInsights' used by this operation is not registered. We are registering for you.
Registration succeeded.
{
  "appId": "eb8a03a-1b1f-40bd-af68-605f43d2177",
  "applicationId": "appi-eklint",
  "applicationType": "web",
  "connectionString": "InstrumentationKey=1ac51431-52f6-4ea3-a130-aa70b486f840;IngestionEndpoint=https://swedencentral-8.in.applicationinsights.azure.com/LiveEndpoint=https://swedencentral.livediagnostics.monitor.azure.com/ApplicationId=eb8a03a-1b1f-40bd-af68-605f43d2177",
  "creationDate": "2026-05-22T10:08:18.391956+00:00",
  "displayname": null,
  "etag": "\"78000b81-0000-7800-0000-6a102b120000\"",
  "flowType": "Bluefield",
  "hockeyAppToken": null,
  "hockeyAppToken": null,
  "id": "/subscriptions/66502789-5701-4eaa-b092-17b9d3d650b0/resourceGroups/rg-TheBookParlour/providers/microsoft.insights/components/appi-eklint",
  "instrumentationKey": "1ac51431-52f6-4ea3-a130-aa70b486f840",
  "location": "swedencentral",
  "name": "appi-eklint",
  "provisioningState": "Succeeded",
  "type": "Microsoft.OperationalInsights/components"
}
```

3. Skapa Application Insights

```
az monitor app-insights component create \
  --app $appInsights \
  --location $location \
  --resource-group $resourceGroupName \
  --workspace $workspace
```

```
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $connectionString = az monitor app-insights component show --app $appInsights --resource-group $resourceGroupName --query connectionString -o tsv
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend> $connectionString
InstrumentationKey=1ac51431-52f6-4ea3-a130-aa70b486f840;IngestionEndpoint=https://swedencentral-8.in.applicationinsights.azure.com/LiveEndpoint=https://swedencentral.livediagnostics.monitor.azure.com/ApplicationId=eb8a03a-1b1f-40bd-af68-605f43d2177
PS C:\Users\emeli\Documents\Projects\InProgress\TheBookParlour\Backend>
```

Connection stringen behövs för att koppla applikationen till Application Insights. Vi sparar den direkt i en variabel.

4. Hämta Connection String

```
$connectionString = az monitor app-insights component show \
  --app $appInsights \
  --resource-group $resourceGroupName \
  --query connectionString \
  -o tsv

# Verifiera att den hämtades
echo $connectionString
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault secret set `
--vault-name $keyVaultName `
--name "ApplicationInsights--ConnectionString" `
--value $connectionString
{
  "attributes": {
    "created": "2026-05-22T12:32:24+00:00",
    "enabled": true,
    "expires": null,
    "notBefore": null,
    "recoverableDays": 99,
    "recoveryLevel": "Recoverable+Purgeable",
    "updated": "2026-05-22T12:32:24+00:00"
  },
  "contentType": null,
  "id": "https://kv-lab-e-kint.vault.azure.net/secrets/ApplicationInsights--ConnectionString/fc9b843bcac8085b958cbe55f4a53",
  "tags": null,
  "managed": null,
  "name": "ApplicationInsights--ConnectionString",
  "tags": {
    "file-encoding": "utf-8"
  },
  "value": "InstrumentationKey=1ac51431-b2fa-4ea3-a130-aa70b486f800;IngestionEndpoint=https://saeedcentral-9.in.applicationinsights.azure.com/liveEndpoint=https://saeedcentral.livediagnostics.monitor.azure.com;/ApplicationId=ebba943a-1b1f-40b1-ae68-685f43d22177"
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az webapp config appsettings set `
--name $webAppName `
--resource-group $resourceGroupName `
--settings ApplicationInsights__ConnectionString="$connectionString"
{
  "name": "ConnectionStrings__DefaultConnection",
  "slotSetting": false,
  "value": null
},
{
  "name": "ApplicationInsights__ConnectionString",
  "slotSetting": false,
  "value": null
}
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> # Verifiera att installationen finns
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az webapp config appsettings list `
--name $webAppName `
--resource-group $resourceGroupName
{
  "name": "ConnectionStrings__DefaultConnection",
  "slotSetting": false,
  "value": "Server=tcp:sql-api-ekint.database.windows.net,1433;Database=TheBookParlourDb;Authentication=Active Directory Managed Identity;Encrypt=True;"
},
{
  "name": "ApplicationInsights__ConnectionString",
  "slotSetting": false,
  "value": null
}
```

5. Lägg till connectionstring som en hemlighet i Azure Key Vault

Detta förutsätter att du har följt guiden för Azure Key Vault.

```
az keyvault secret set `
--vault-name $keyVaultName `
--name "ApplicationInsights--ConnectionString" `
--value $connectionString
```

Alternativ (ej rekommenderat)

Om man inte har Azure KeyVault kan man ha connection string i appsettings, men det är inte lika säkert.

Sätt Connectionstring som appsetting och verifiera att den skapades.

```
az webapp config appsettings set `
--name $webAppName `
--resource-group $resourceGroupName `
--settings ApplicationInsights__ConnectionString="$connectionString"

# Verifiera att installationen finns
az webapp config appsettings list `
--name $webAppName `
--resource-group $resourceGroupName
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az keyvault secret show `
--vault-name $keyVaultName `
--name "ApplicationInsights--ConnectionString" `
--query value
{
  "value": "InstrumentationKey=1ac51431-b2fa-4ea3-a130-aa70b486f800;IngestionEndpoint=https://saeedcentral-9.in.applicationinsights.azure.com/liveEndpoint=https://saeedcentral.livediagnostics.monitor.azure.com;/ApplicationId=ebba943a-1b1f-40b1-ae68-685f43d22177"
}
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

```
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend> az role assignment list `
--assignee $userId `
--scope $keyVaultId `
-o table
Principal Role Scope
-----
azurejensen148@studentindjensen.omeicrosoft.com Key Vault Secrets Officer /subscriptions/96582789-5741-4eaa-bb92-1768d3d604b9/resourceGroups/rg-TheBookParlour/providers/Microsoft.KeyVault/vaults/kv-lab-e-kint
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
PS C:\Users\emeli\Documents\ProjectsInProgress\TheBookParlour\Backend>
```

6. Hämta hemligheten

För att kontrollera att hemligheten lagrades korrekt i Key Vault, att du har rättigheter att läsa den samt att värdet är korrekt.

```
az keyvault secret show `
--vault-name $keyVaultName `
--name "ApplicationInsights--ConnectionString" `
--query value `
-o tsv
```

7. Verifiera dina roller

```
az role assignment list `
--assignee $userId `
--scope $keyVaultId `
-o table
```

Fas 2: Integrera Application Insights i .NET-applikationen

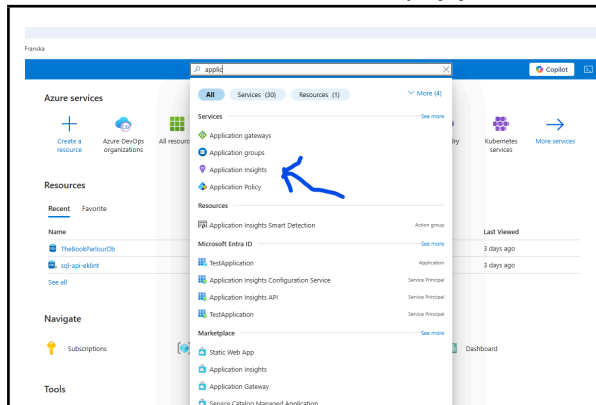
1. Installera nuget-paket Visual Studio (Package Manager Console)

```
dotnet add package
Microsoft.ApplicationInsights.AspNetCore
NetCore
```

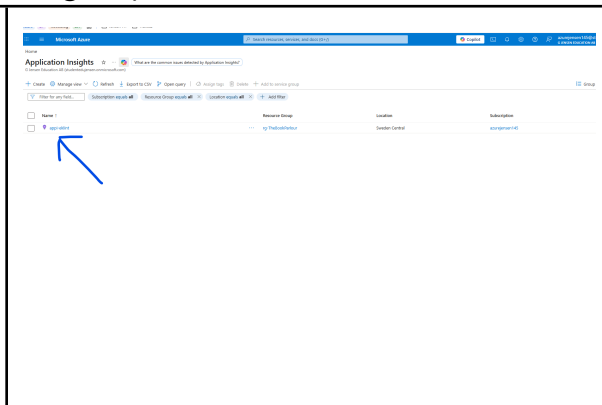
2. Lägg till i program

```
// Connection string hämtas
automatiskt från konfigurationen
builder.Services.AddApplicationIn
sightsTelemetry();
```

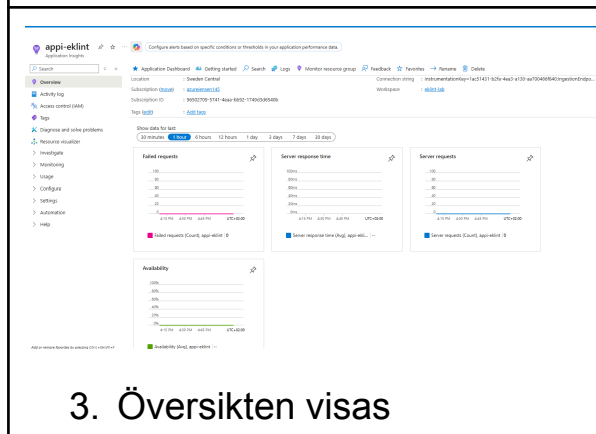
Fas 3: Azure Portalen (Application Insights)



1. Sök efter application Insights



2. Klicka på den du vill visa.



3. Översikten visar

